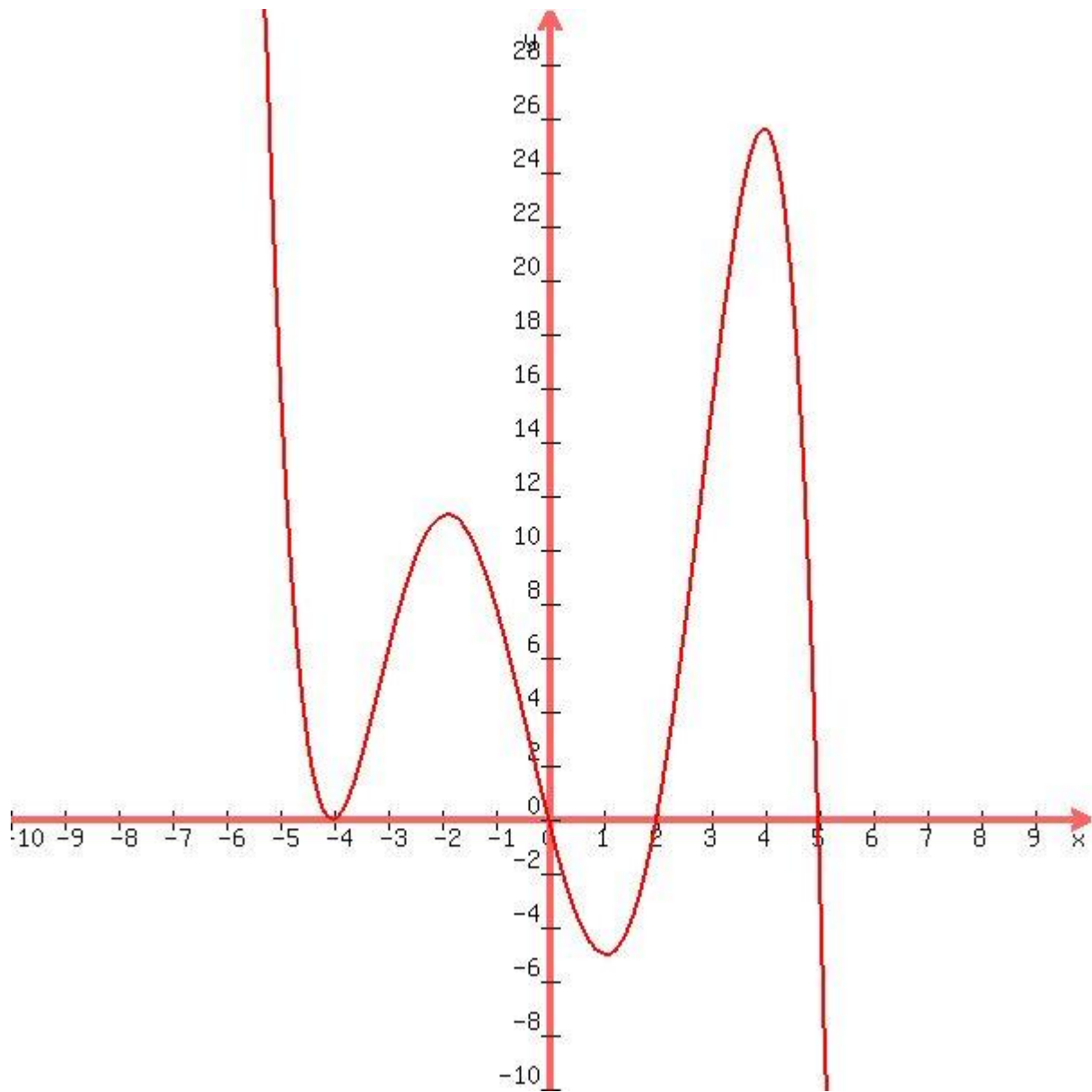


Assignment-01

1. According to the following figure find:

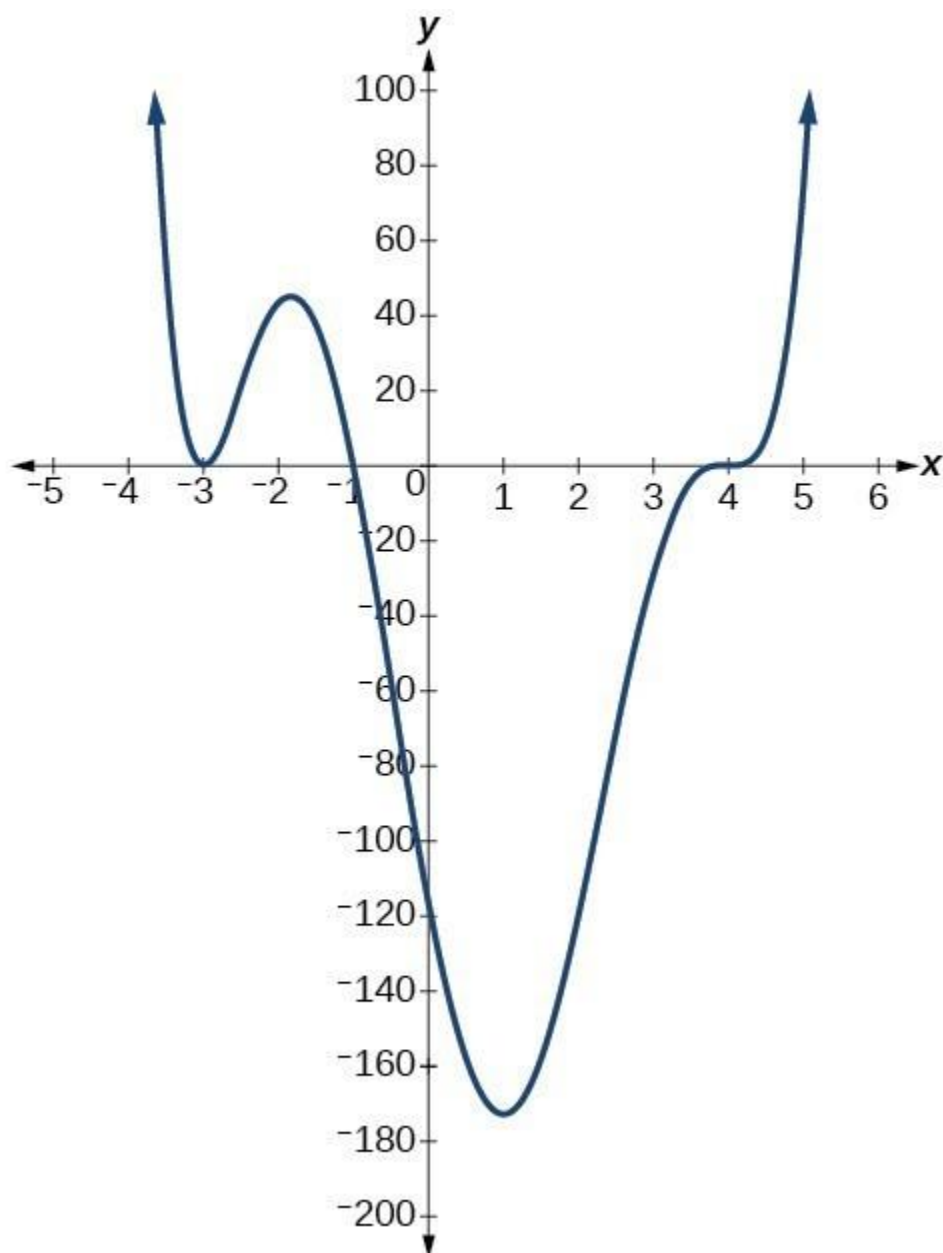
- (i). The intervals at which the graph of function increases/decreases.
- (ii). Critical points.
- (iii). Maximum/minimum values of the given function.
- (iv). The intervals at which the graph of function concave up/down.
- (v). Inflection points.



Assignment-01

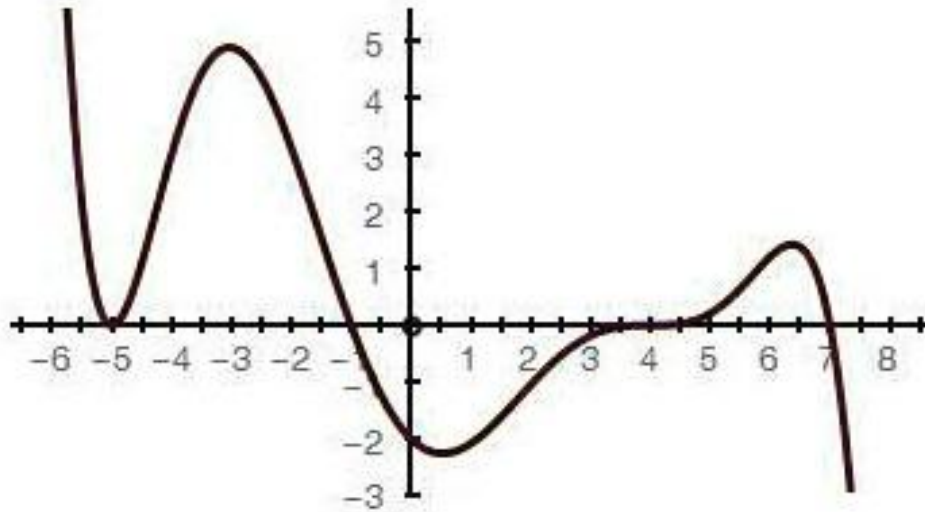
2. According to the following figure find:

- (i). Find the signs of $f'(x)$ and $f''(x)$.
- (ii). What are the critical and inflection points?

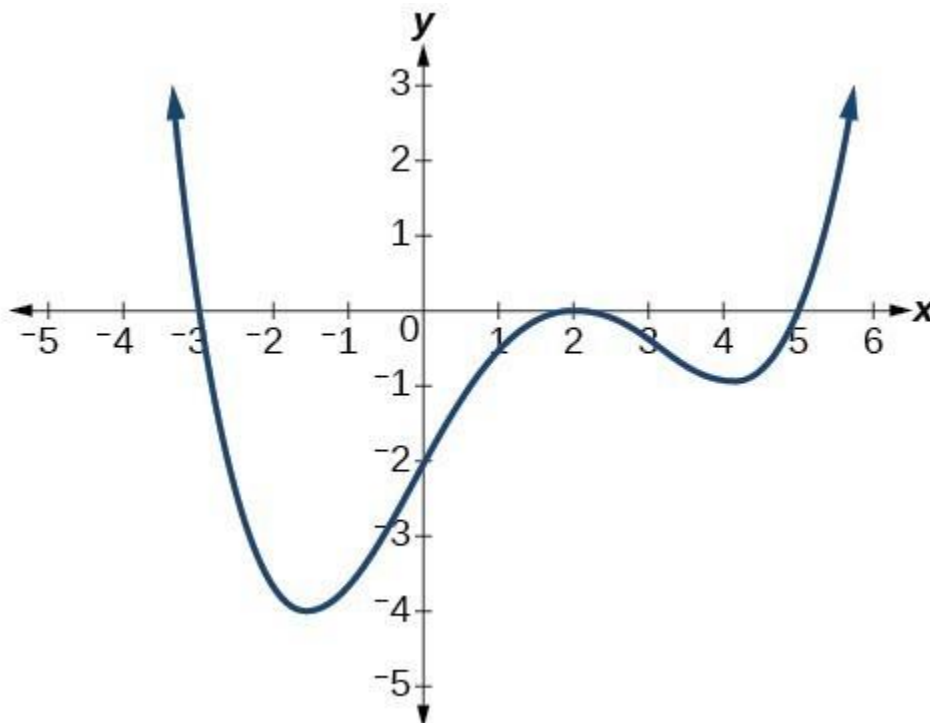


Assignment-01

3. According to the following figure find the values of x at which the $f'(x)$ and $f''(x)$ are zero. Discuss your findings.



4. According to the following figure, find the nature (even/odd multiplicities) of the roots.



Assignment-01

5. Consider $f(x) = 2x^3 + 3x^2 - 36x$ to find the maximum and minimum by the first derivative test. Also, verify by the second derivative. Find the points at those $f(x)$ changes the shape?
6. For the polynomial $f(x) = (x - 2)^3(x^2 + 1)(1 + 2x)^4$ find the nature of the graph at the position of the roots.
7. Find the nature of the polynomial $f(x) = x^3 + 4x^2 - 3x - 18$ and hence sketch it.